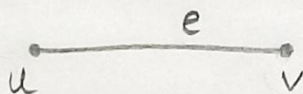


Unit - IV

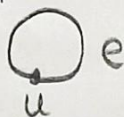
Graph Theory

A graph $G = (V(G), E(G))$ consists of 2 finite sets where $V(G)$ is the vertex set of G which is a non-empty set of elements called vertices. It is denoted by V and $E(G)$ is the edge set of G which is possibly a non-empty set of elements called edges. E is denoted by E such that each edge in G is assigned a pair of vertices called end-vertices.

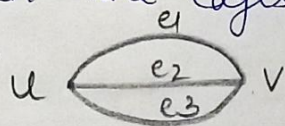
Note:- ① If " e " is an edge with end vertices u & v then e is said to join u & v .



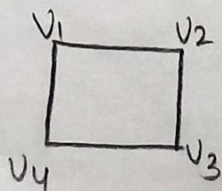
② In a graph G with edge e and same end vertices is called a loop or any vertex is joined itself by an edge is called a loop or self loop.



③ In a graph G , 2 or more edges having the same end vertices, then the edges are called parallel edges.



④ In a graph G , a vertex which is not the end of any vertex is called isolated vertex.



v_5

v_5 - isolated vertex

⑤ Adjacent vertices - 2 vertices joined by an edge are called adjacent vertices. If 2 edges u & v are said to be adjacent if there exists an edge b/w them.

⑥ Simple graph - G is said to be a simple graph if it has no parallel edges and no self-loops.



simple



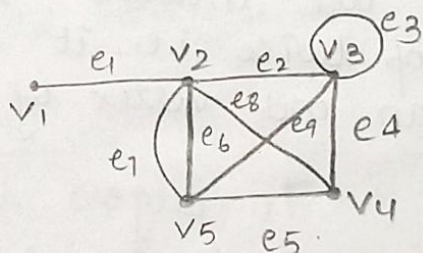
self loop

parallel edge

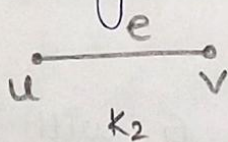


not simple

eg:- Graph:-



⑦ A simple graph G is said to be a complete graph if every pair of distinct vertices are joined by an edge.



K_2



K_3

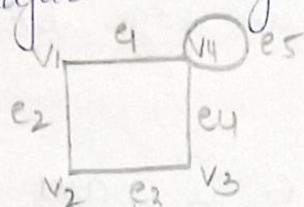


K_4

The complete graph of n vertices is denoted by K_n . Maximum number of edges in a complete graph of n vertices is $\frac{n(n-1)}{2}$ edges.

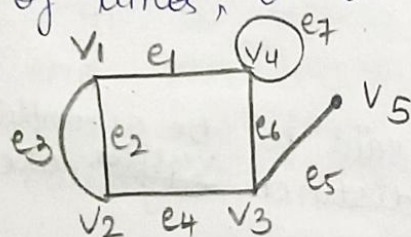
⑧ An edge e of a graph G is said to be incident with a vertex v if v is an end vertex of e . In this case, we also say that v is incident with e — Incident edges

⑨ Adjacent edges - Two edges e_1 & e_2 which are incident with a common vertex are called adjacent edges.



Note :- An edge is adjacent with either 1 or 2 vertices. 1 vertex if it is a self loop whereas a vertex may be incident with a finite ~~ed~~ no. of edges including zero (isolated).

Degree of Vertex :- Let V be the vertex set of graph G . The degree of $v \in V$ is the no. of edges of G that are incident with v counting each loop twice i.e., it is the no. of times v is an end vertex of an edge.



7 - edges

$d(v)$ - degree

$$d(v_1) = 3$$

$$d(v_2) = 3$$

$$d(v_3) = 3$$

$$d(v_4) = 4$$

$$d(v_5) = 1$$

$$\Rightarrow \text{Total} = 14$$

$$= 2 \times (7)$$

no. of edges

Theorem (1) :- For a graph G with e edges and n vertices $v_1, v_2, \dots, v_n$

$$\text{then } \sum_{i=1}^n d(v_i) = 2e.$$

Proof :- In a graph G , each edge has 2 end vertices so it contributes precisely 2 to the sum of degrees of all vertices i.e., when the degrees of all vertices are

added, all edges are counted twice.

$$\text{i.e., } \sum_{i=1}^n d(v_i) = 2e$$

Odd Vertices & Even Vertices

A vertex of a graph G is called odd or even depending on whether its degree is odd or even.

Eg:- In the previous example.

$v_1, v_2, v_3, v_5 \rightarrow$ odd vertex

$v_4 \rightarrow$ even vertex

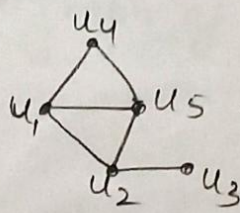
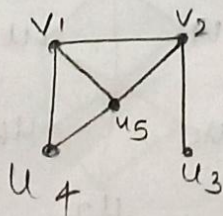
Note:- In a graph G , there are even no. of odd vertices.

Isomorphic

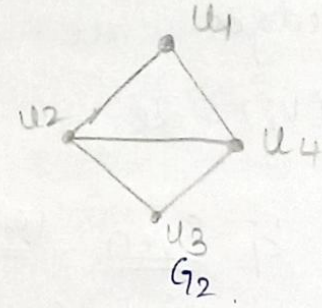
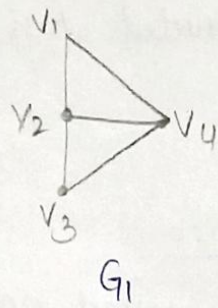
A graph $G_1 = (V, E)$ is said to be isomorphic to $G_2 = (U, F)$ if there is one to one correspondence between the vertex set U & V & edge set E & F in such a way that $e_1 \in E$ is an edge between v_1, v_2 in G_1 , then the corresponding edge $f_1 \in F$ in G_2 with u_1, u_2 as end vertices. Such a pair of correspondence is called as graph isomorphism.

Note :- Two graphs G_1 & G_2 are said to be isomorphic if they must have same number of edges & vertices.

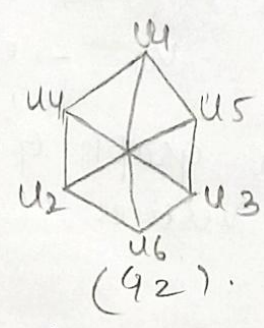
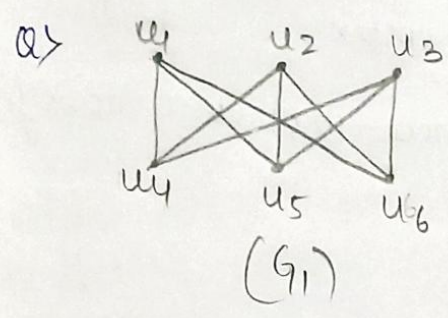
Eg:- G_1 & G_2 are isomorphic.



(Eg:-2



Here G_1, G_2 contain 4 vertices & 5 edges and there is one to one correspondence between vertices & edges. So, G_1 & G_2 are isomorphic.



Name vertices in G_2 such that G_1 & G_2 are isomorphic.

A> $u_1 \xrightarrow{\text{all}}$ except u_2, u_3

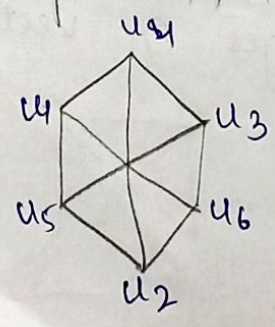
$u_2 \xrightarrow{\text{all}}$ except u_1, u_3

$u_3 \xrightarrow{\text{all}}$ except u_1, u_2

$u_4 \xrightarrow{\text{all}}$ except u_5, u_6

$u_5 \xrightarrow{\text{all}}$ except u_4, u_6

$u_6 \xrightarrow{\text{all}}$ except u_4, u_5



Regular Graph

A graph G in which all the vertices are of equal degree is said to be a regular graph.

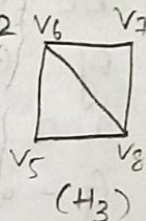
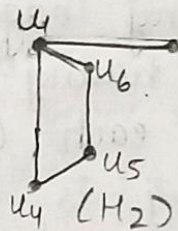
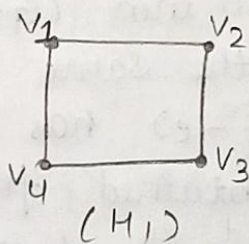
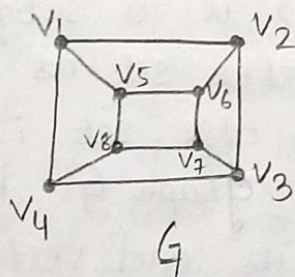
Eg:- In the above question G_1 & G_2 are regular.

Subgraph

Let H be a graph with vertex set $V(H)$ and edge set $E(H)$ and G be another graph with vertex set $V(G)$ & edge set $E(G)$.

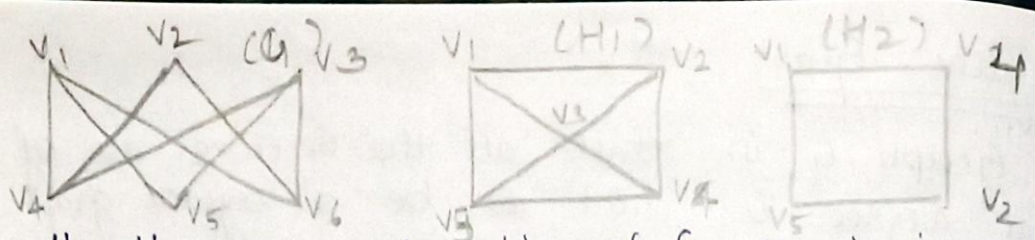
We say that H is a subgraph of G if $V(H)$ is contained in $V(G)$ i.e., $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$.

Eg:- Check if H_1 & H_2 are subgraphs of G



A> H_1, H_2 are subgraphs of G
 H_3 is not graph of G .

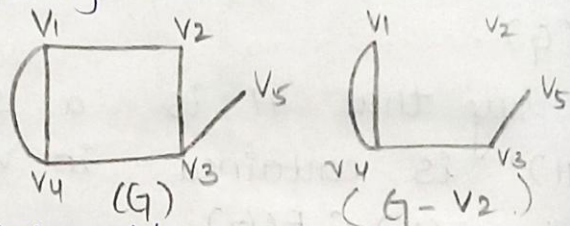
Note:- Any graph isomorphic to subgraph of G is also referred as subgraph of G .



H_1, H_2 are subgraphs of G . H_2 is isomorphic to H_1 .

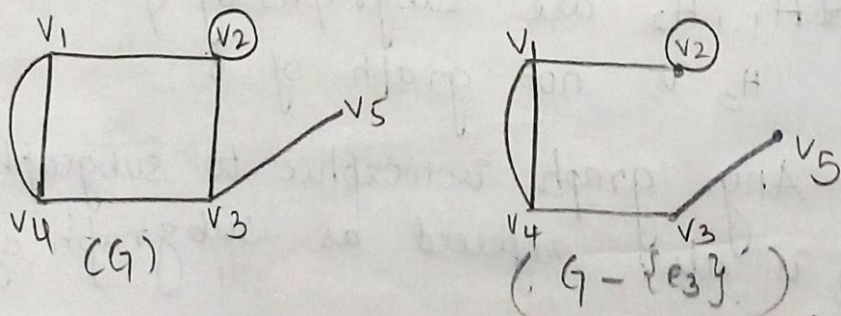
Vertex deleted subgraph

In a graph G , which has at least 2 vertices then for any vertex $v \in G$, $G - v$ denotes subgraph of G with vertex set $V - v$ & whose edges are all those of G which are not incident with v i.e., $G - v$ is obtained by removing vertex v and all edges of G which have v as an end vertex.



Edge deleted subgraph

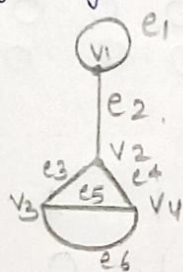
If $G = (V, E)$ is a graph and $e \in E$ be any edge then $(G - E)$ is a subgraph of G having the same vertex set as G and each $(E - e)$ has its edge set i.e., $G - \{e\}$ is obtained from graph G by removing edge e but not its end vertices.



Matrix Representation of Graphs

>> Adjacency Matrix

Let G be a graph with n vertices $v_1, v_2, \dots, v_n$. The adjacency matrix of G with n vertices of G is a $n \times n$ matrix $A(G) = [a_{ij}]$, ($i = 0, 1, 2, \dots, n$) where (i, j) th entry a_{ij} is the number of edges joining the vertex v_i to vertex v_j .



$$A(G) = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 2 & 0 \end{bmatrix} \end{matrix}$$

Note :- i) In the adjacency matrix $A(G)$, $a_{ij} = a_{ji}$ i.e., adjacency matrix is symmetric.

- ii) If G has no loops then all the entries in the main diagonal are zero's.
- iii) If there are no parallel edges, then all the entries of $A(G)$ are either 0 or 1's.

>> Incidence Matrix

Let G be a graph with n vertices $v_1, v_2, \dots, v_n$ and m edges $e_1, e_2, \dots, e_m$. The incidence matrix of G is a $n \times m$ matrix given by

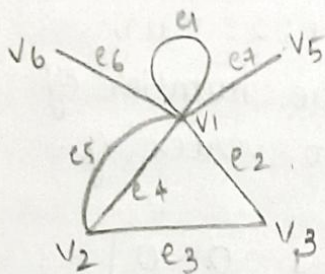
$$M[G] = [m_{ij}] \quad \begin{matrix} i = 1, 2, \dots, n \\ j = 1, 2, \dots, m \end{matrix}$$

m_{ij} - no. of times the vertex v_i is incidence on the edge e_j .

$$m_{ij} = \begin{cases} 0 & \text{if } v_i \text{ is not an end-vertex of } e_j \\ 1 & \text{if } v_i \text{ is an end vertex of non-loop } e_j \\ 2 & \text{if } v_i \text{ is an end vertex of loop } e_j \end{cases}$$

Eg:-

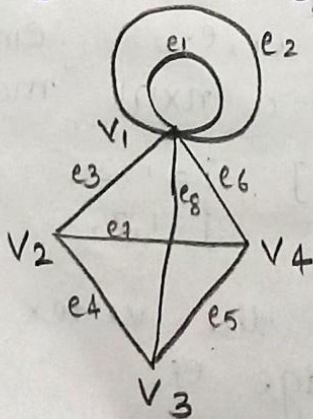
Q5 Find $A(G)$ & $M(G)$



$$A(G) = \begin{bmatrix} 1 & 2 & 1 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M(G) = \begin{bmatrix} 2 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

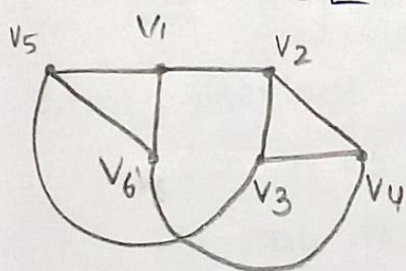
Q6. $A(G) = \begin{bmatrix} 2 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ Draw Graph & find $M(G)$.



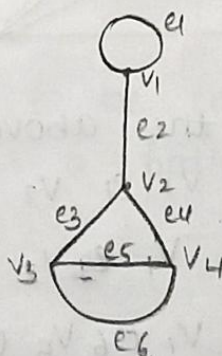
$$M(G) = \begin{bmatrix} 2 & 2 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

Q. Draw graph from $M(G)$.

$$M(G) = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 & e_9 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \\ v_6 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$



Q. Find $M(G)$ for



$$M = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

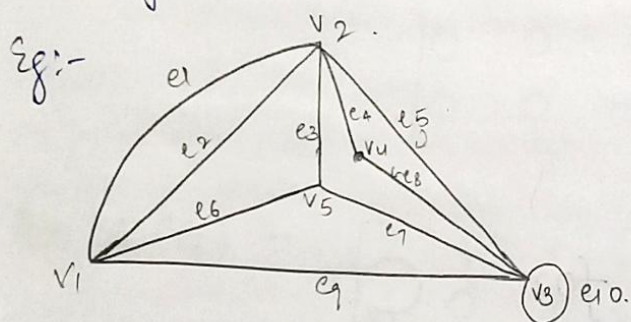
Walk, Path, Trial, Cycles :- A walk in a graph is a finite sequence

$$W = v_0 e_1 v_1 e_2 v_2 \dots v_{k-1} e_k v_k$$

where terms are alternatively vertices & edges such that $1 \leq i \leq k$, the edge e_i has end vertices v_{i-1} & v_i

We can also say that the above walk w is a walk from v_0 to v_k . The vertex v_0 is called the origin of walk and vertex v_k is called terminus of the walk & the vertices $v_1, v_2, \dots, v_{k-1}$ are called the internal vertices of the walk.

In a walk w , the no. of edges is called the length of the walk (i.e., k is length here).



In the above graph G

$$w_1 = v_1 e_1 v_2 e_5 v_3 e_9 v_3 e_5 v_2 e_3 v_5$$

$$w_2 = v_1 e_2 v_2 e_2 v_1 e_1 v_2$$

$$w_3 = v_1 e_6 v_5 e_3 v_2 e_4 v_4 e_8 v_3 v_9 v_1$$

$$= v_1 v_5 v_2 v_4 v_3 v_1$$

Here w_1, w_2, w_3 are of lengths 5, 3, 5 respectively.

In a graph G , if the origin & terminals are same, then it is closed walk otherwise it is an open walk.

In above example

$w_1, w_2 \rightarrow$ open walk

$w_3 \rightarrow$ closed

Trial :- A walk w is said to be trial if which no edge is repeated.

In above examples, w_3 - trial $\checkmark$

w_2, w_1 - trial $\times$

Path :- A walk is said to be a path if no vertex is repeated.

$\gg$ Every path is a trial but every trial is not a path.

From previous example.

$w_4 = v_1 e_2 v_2 e_5 v_3 e_8 v_4$.

$\rightarrow$ path & trial.

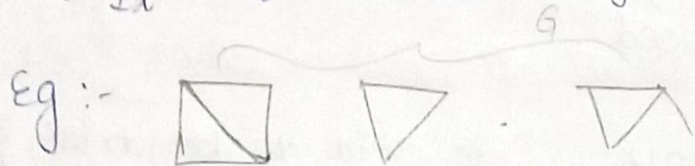
Connected Graph Vertices

A vertex u is said to be connected to a vertex v , if there is a path from u to v in G .

Connected Graph

A Graph G is said to be connected if there exists a path between every pair of vertices otherwise Graph G is disconnected

Component :- If a graph G is disconnected then the number of connected subgraph of G is called component. It is denoted by $w(G)$.



$w(G) = 4$ i.e., in G no. of components is 4.

Cycle :-

A non-trivial closed trail in a graph G is called a cycle if its origin & internal vertices are distinct. In other words, the closed trail $C = v_1 v_2 \dots v_n v_1$ is a cycle if C has at least one edge and $v_1 v_2 v_3 \dots v_n$ are all distinct.

The no. of edges in a cycle is called the length of the cycle.

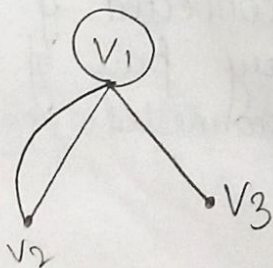
In general, k -cycle means cycle of length k .

1-cycle - self loop.

2-cycle - has

3-cycle - often called a triangle.

Eg :- Consider the following graph (G)



$\Rightarrow$ Adjacency matrix $A(G)$ is

$$A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \end{matrix} & \begin{bmatrix} 1 & 2 & 1 \\ 2 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

Now

$$A^2 = A \times A = \begin{bmatrix} 6 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} = [b_{ij}]$$

6 different walks.

Here b_{ij} indicates no. of different walks of length 2 from vertex v_i to vertex v_j .

Euler Trial

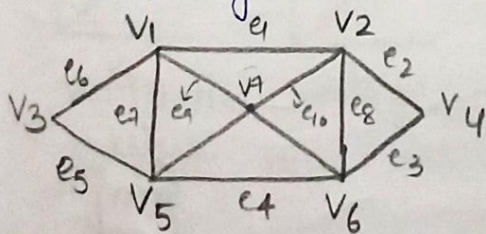
A trial in a graph G is called an Euler trial if it includes every edge of G .

$\therefore$ Trial is said to be euler if each edge is there exactly once. Eg:- 

Tour :- A closed walk of G which includes every edge of G ^{atleast} ~~exactly~~ once.

Euler Tour :- A euler tour of graph G which includes every edge of G exactly once.

Euler Graph :- A graph G is said to be euler if it has an euler tour.

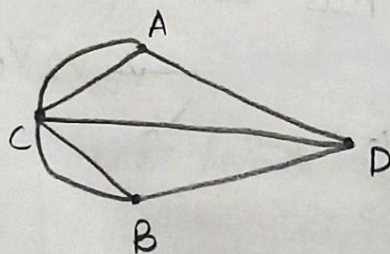
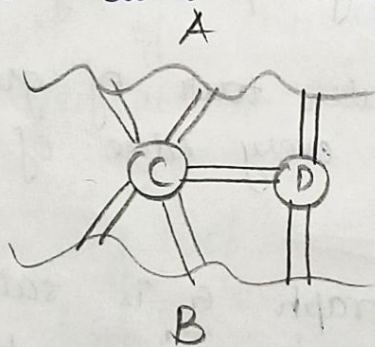


$V_5 V_2 V_6 V_1 V_2 V_4 V_6 V_5$
 $V_3 V_1$
 V_5

↓
euler tour

Konigsberg Bridge Problem

Early in the 18th century, the Euler considered problem of 7 Konigsberg Bridges. 2 islands C and D are formed a river in the Persian city Konigsberg which were formed connected to each other to banks A & B with 7 bridges as shown below. The problem was to start at any one of the 4 land areas A, B, C, D and walk over each of the 7 bridges exactly one & return to starting point without swimming across river. Euler represented this by means of graph as shown below where vertices are land areas & bridges are the edges. It is not possible to walk across each of the bridges exactly once & return to starting point because graph has no Euler trail.



→ not an Euler graph

Note: A connected graph G is euler if and only if every vertex is even.

Hamiltonian path — It is a path of graph G which contains every vertex of G .

→ Since no vertex is repeated in path, Hamiltonian path in G contains every path of G exactly once.

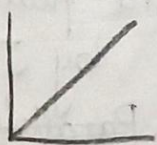
Hamiltonian cycle/circle — A hamiltonian cycle in graph G is a cycle which contains every vertex of G .

→ Since no vertex is repeated apart from origin and terminus, a hamiltonian cycle with initial vertex v contains every other vertex of G exactly once and then ends at the vertex v .

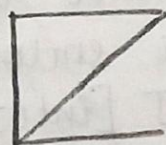
Hamiltonian Graph — Graph should have hamiltonian cycle.

» By simply deleting last edge of an hamiltonian cycle, we get a hamiltonian path.

» A non-hamiltonian graph may possess a hamiltonian path



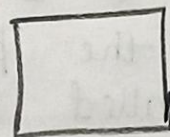
G_1



G_2



G_3



G_4

G_1 has no hamiltonian path

G_2, G_3 have hamiltonian path

G_4 has hamiltonian cycle.

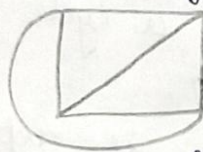
Plane Graphs and Planar Graphs

→ A plane graph is a graph drawn on the plane in such a way that any pair of edges ~~meet~~ only at their end vertices if they meet at all.

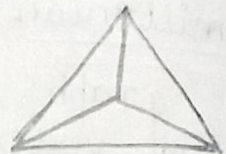
→ A planar graph is a graph which is isomorphic to plane graph i.e., can be re-drawn as plane graph.



(G₁)



(G₂)



(G₃)

Here all graphs are planar but G_1 is not a plane graph.

Also G_2 & G_3 are redrawings of G_1 .

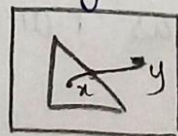
Jordan Curve — A Jordan curve in a plane is a continuous non-self intersecting curve whose origin & terminus coincide.

Note :- i) Any Jordan Curve divides the plane into 2 regions called the interior & exterior region.

ii) If J is the Jordan Curve in the plane, then the part of plane enclosed by J is called interior of J [$\text{int}(J)$].

Similarly part of plane ~~lying~~ outside J is called exterior of J [$\text{ext}(J)$].

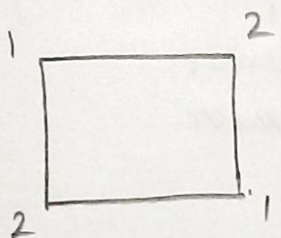
iii) If J is a Jordan curve & if x is a pt in $\text{int}(J)$ & y in $\text{ext}(J)$, then any point on line joining x & y must meet J i.e., Jordan curve at some point



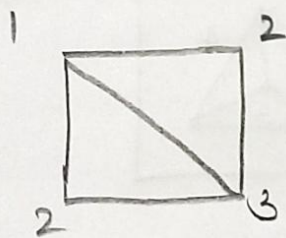
Graph Colouring (Vertex Colouring)

Let G be a graph which is simple & has no self-loops. In the graph G we colour the vertices with different colours in such a way that 2 adjacent vertices can't have same colour.

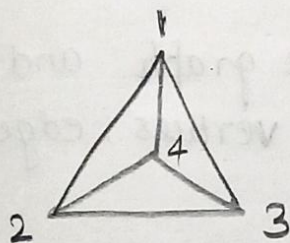
→ Chromatic number - The minimum number of colours that are used to colour a graph is called chromatic number. Denoted by $\chi(G)$.



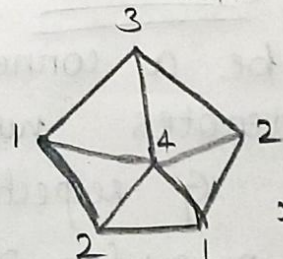
$$\chi(G) = 2$$



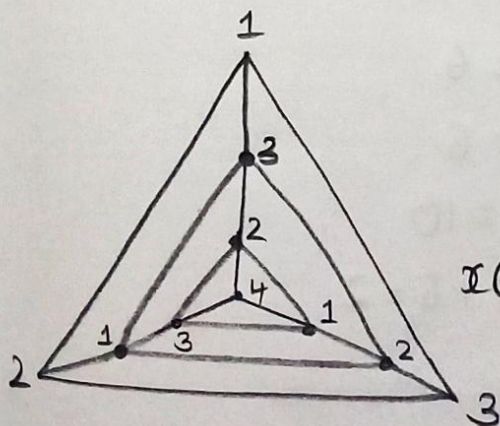
$$\chi(G) = 3$$



$$\chi(G) = 4$$



$$\chi(G) = 5$$



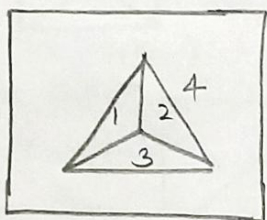
$$\chi(G) = 4$$

Faces of a Graph

A plane graph G partitions the plane into number of regions called faces of graph. It is denoted by f or $f(G)$.

Note: i) Any plane graph has exactly 1 exterior face.

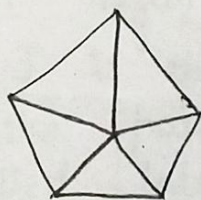
ii) Any other face bounded by a closed walk in the graph is called interior face.



This graph has 4 faces
3 - interior
1 - exterior

Euler's Formula

Let G be a connected plane graph and let n, e, f denotes number of vertices, edges & faces of G respectively then $n - e + f = 2$



$$f = 6$$

$$n = 6$$

$$e = 10$$

$$6 - 10 + 6 = 2$$